MIL-P-48819(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

PLUGS, ELECTRICAL, TRAINING, FOR M74 WARHEAD SECTION

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for four types of special training plugs, to complete electrical circuits, which are used for training on the Nike Hercules warhead section.

Type I - Training Plug, Electrical, Surface-to-Air
Type II - Training Plug, Electrical, Surface-to-Surface
Type III - Training Plug, Electrical, Safe
Type IV - Training Plug, Electrical, Firing

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8801514 - Connector
8801718 - Connector
8806600 - Bag, Packing
8837412 - Packing

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8846008 - Plug, Safe
8846009 - Plug, Firing
8846010 - Plug, Surface-to-Air
8846011 - Plug, Surface-to-Surface
8846373 - Warhead Section Assembly - Packing Drawing
EL-8855805 - Equipment List

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The electrical plugs shall comply with all requirements specified on the drawing (dwg) 8846008, 8846009, 8846010 and 8846011, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Continuity.

3.4.1 Electrical Plug, Safe. The continuity of the Electrical Plug, Safe shall be as specified in Table I when tested as specified in 4.4.1.

Table I

From	To
A	M
E	F
F	N
O	G

3.4.2 Electrical Plug, Firing. The continuity of the Electrical Plug, Firing shall be as specified in Table II when tested as specified in 4.4.1.

Table II

From	To
H	I
K	L
M	N
N	P
P	O

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3.4.3 Electrical Plug, Surface-to-Air. The continuity of the Electrical Plug, Surface-to-Air shall be as specified in Table III when tested as specified in 4.4.1.

Table III

From	To
U	M
U	f

3.4.4 Electrical Plug, Surface-to-Surface. The continuity of the Electrical Plug, Surface-to-Surface shall be as specified in Table IV when tested as specified in 4.4.1.

Table IV

From	To
U	B
U	b

3.5 Insulation resistance.

3.5.1 Electrical Plug, Safe. The insulation resistance of the Electrical Plug, Safe, when subjected to a potential of 500 + 50 volts direct current (vdc), shall not be less than 20 megohms between the circuits specified in Table V when tested as specified in 4.4.2.

Table V

From	To	Except pins
Connector Shell	All pins	--
A	All other pins	M
E	All other pins	F & N
O	All other pins	G

3.5.2 Electrical Plug Firing. The insulation resistance of the Electrical Plug, Firing when subjected to a potential of 500 + 50 vdc, shall not be less than 20 megohms between the circuits specified in Table VI when tested as specified in 4.4.2.

Table VI

From	To	Except pins
Connector Shell	All pins	---
H	All other pins	I
K	All other pins	L
M	All other pins	N, P and O

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3.5.3 Electrical Plug, Surface-to-Air. The insulation resistance of the Electrical Plug, Surface-to-Air, when subjected to a potential of 500 ± 50 vdc, shall not be less than 20 megohms between the circuits specified in Table VII when tested as specified in 4.4.2.

Table VII

From	To	Except pins
Connector Shell M	All pins	---
	All other pins	I, N, f, and U

3.5.4 Electrical Plug, Surface-to-Surface. The insulation resistance of the Electrical Plug, Surface-to-Surface, when subjected to a potential of 500 ± 50 vdc, shall not be less than 20 megohms between the circuits specified in Table VIII when tested as specified in 4.4.2.

Table VIII

From	To	Except pins
Connector Shell B	All pins	---
	All other pins	S, b, N and U

3.6 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.7.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier

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is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwgs 8846008, 8846009, 8846010 and 8846011) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection samples shall be subjected to the inspections and tests specified in Table IX. Tests shall be conducted in order specified in Table IX.

Table IX

First article inspection and tests

Inspection or Tests	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity (see 3.4)		X
3. Insulation resistance (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table X and Table XI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table X
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table XI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

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4.3.2.1.1 Connector, prior to assembling cap (see dwgs 8801514 and 8801718).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Outside diameter	Gage
102.	Contact damaged	Visual
103.	Crack in insulation material	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.1.2 Plugs, Electrical (see dwgs 8846008, 8846009, 8846010 and 8846011).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Coupler thread damaged (as applicable)	Gage/Visual
102.	Plug damaged or any part missing	Visual
103.	Streamer, color and length incorrect (as applicable)	Visual
104.	Lacquer missing	Visual
105.	Marking incorrect, mislocated, or unidentifiable	Visual
106.	Cap loose	Manual
107.	Solder loose, inadequate, or cold solder joint (see 3.6)	Visual
*108.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pin)	Visual/Gage
109.	Any pin bent	Visual
110.	Pin missing (as evidenced by empty hole in insert)	Visual
111.	Foreign matter on any pin or in any pin hole unless removable	Visual
112.	Portion of connector broken away (minor chips, scratches, burrs or gouges that do not affect the mating are not objectionable)	Visual

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113. Gap between coupling ring and cap (8846009), 8846010 & 8846011).....Gage
114. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:

201. Total length, max.....Gage
202. Evidence of poor workmanship (see 3.7).....Visual
203. Marking misleading, missing or unidentifiable.....Visual
204. Coating inadequate.....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing not in accordance with MIL-STD-1169	Visual
	102. Packing not in accordance with applicable drawing	Visual
	103. Marking not in accordance with dwg 8806600	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.7)	Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4.1) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. Continuity	Major
b. Insulation resistance	Major

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None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity and insulation resistance. Each type plug in the lot shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.4 and 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8855805). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity. The circuit resistance of the electrical plug shall not exceed 0.01 ohms, when tested utilizing an ohmeter.

4.4.2 Insulation resistance. The insulation resistance of the electrical plugs shall be tested utilizing a meter capable of reading in megohms.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The plugs shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The plugs shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

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5.2 Packing.

5.2.1 Shipment with trainer. When shipped as a component of the trainer, the plugs shall be installed in accordance with dwg 8846373.

5.2.2 Spares.

5.2.2.1 Level A and C. The plugs when shipped as spares shall be packed in accordance with P/N 8837412 or dwg 8806600.

5.3 Marking. Marking shall be in accordance with dwg 8806600.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.1).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase description -PA-PD-2123 Rev. 1 dated 27 December 1963.

Custodian:
Army - PA

Preparing activity:
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